BUSINESS INSIGHT REPORT

AIML Transactions and Fraud Analysis Dataset

Interactive Power BI Dashboard Analysis



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Data source	AIML Transactions & Fraud Analysis (Power BI report: accounts, alerts and transactions tables)

2. Introduction

2.1 About the AIML Transactions and Fraud dataset

The AIML Transactions and Fraud dataset is an interactive Power BI report that tracks financial transactions across a base of 10,000 accounts. The data model contains three tables: accounts, alerts and transactions. Across the full dataset, the dashboard summarises about 1.32 million transactions with a combined value of 153.48 billion, an average transaction value of 115.99K, and 1,719 transactions flagged as fraudulent. The accounts together hold a total initial balance of 2.80 million.

Each transaction carries an IS_FRAUD flag (True or False) and a TIMESTAMP step that runs from 0 to about 200. The alerts table classifies suspicious activity into two ALERT_TYPE patterns, cycle and fan_in, which are typical anti-money-laundering (AML) typologies. Total Transactions, Fraudulent Transactions, Total Transaction Amount and Total Accounts are the core measures that drive the visuals.

2.2 Purpose of the dashboard

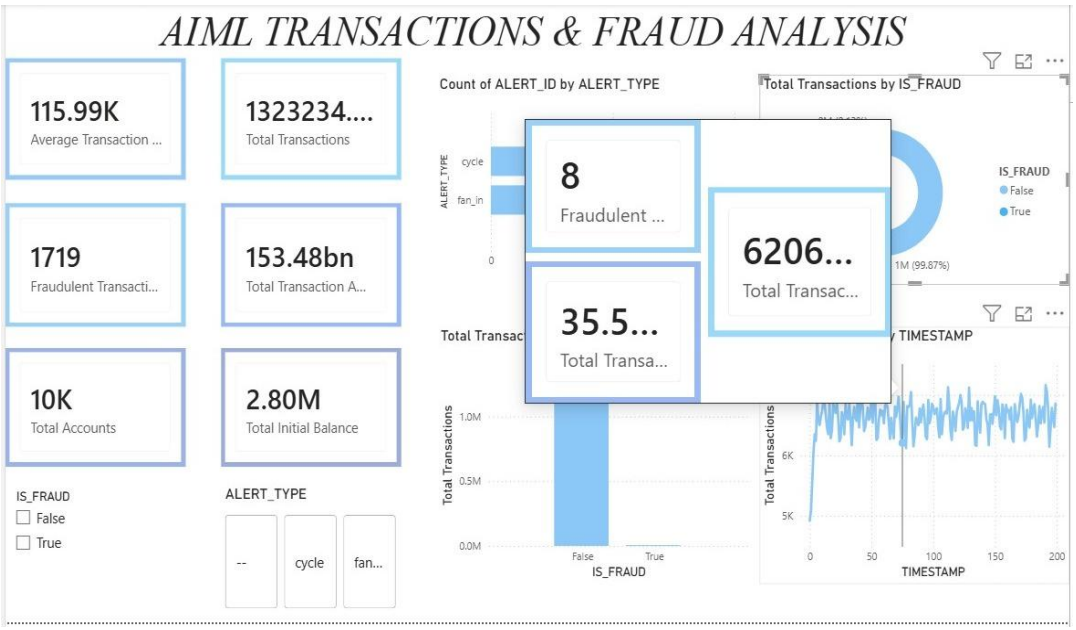
The dashboard gives an analyst one place to monitor transaction volume, transaction value, fraud counts, alert types and the transaction trend over time. The IS_FRAUD and ALERT_TYPE slicers let a user narrow every visual to fraud or non-fraud records, or to a specific alert pattern, without building a new report. A custom Tooltip page adds a quick three-measure summary when the user hovers over a point on the trend line.

2.3 Purpose of this Business Insight Report

The dashboard shows what the data says; this report explains what it means. It interprets each KPI and chart, separates observations (what the data shows) from insights (what the observation means for the business), and turns those insights into practical recommendations for fraud monitoring, alert review and reporting. All numbers come from the supplied screenshots, and values read from chart shapes are marked as estimates.

3. Dashboard Overview

3.1 Dashboard screenshot



3.2 What the dashboard answers

The dashboard answers five everyday monitoring questions: How many transactions are processed and what is the average one worth? How many of them are fraudulent? How large is the account base and how much money did it start with? Which alert types are raised most often? How does transaction volume move over time?

3.3 Key components

KPI cards: six headline measures: Average Transaction Value, Total Transactions, Fraudulent Transactions, Total Transaction Amount, Total Accounts and Total Initial Balance.

- Count of ALERT_ID by ALERT_TYPE: a horizontal bar chart comparing the number of alerts raised for the cycle and fan_in patterns.
- Total Transactions by IS_FRAUD (donut): a donut chart showing the share of fraud (0.13%) against non-fraud (99.87%) transactions.
- Total Transactions by IS_FRAUD (column): a column chart showing the same split as volume (about 1M+ non-fraud against a barely visible fraud column).

- Total Transactions by TIMESTAMP: a line chart of transaction volume for each timestamp step from 0 to about 200.
- Slicers: an IS_FRAUD slicer (False / True) and an ALERT_TYPE slicer (cycle, fan_in and a blank option) that filter every visual at once.
- Tooltip page: a hidden page that shows Fraudulent Transactions, Total Transactions and Total Amount when the user hovers over a chart point.

3.4 Interactive filters and tooltip

The two slicers control every visual simultaneously, and clicking a chart element also cross-filters the rest of the page. For example, selecting True in the IS_FRAUD slicer isolates the 1,719 fraudulent transactions, and selecting an alert type shows only that pattern. The tooltip page, shown below in design view, is displayed as a floating card when a user hovers over a point on the line chart.

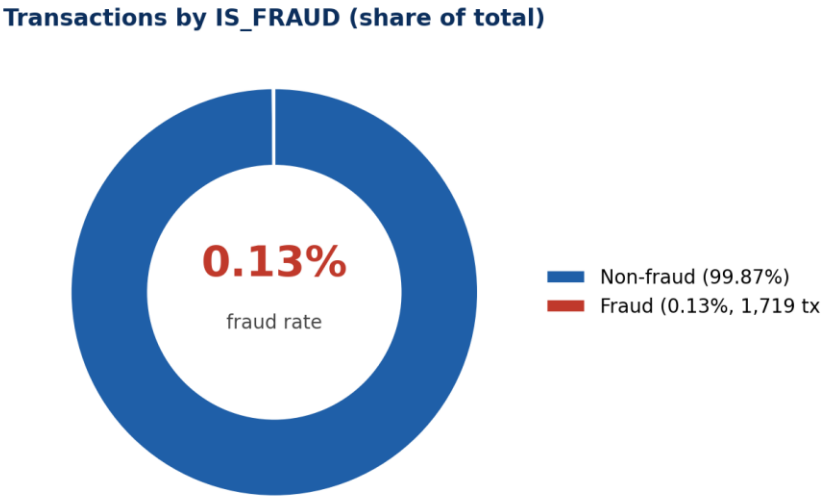


4. KPI Analysis

4.1 Transaction volume and average value

The dashboard records about 1.32 million transactions (the card shows 1323234 and is truncated on screen) with an average value of 115.99K each. Multiplying the two gives roughly 153.5 billion, which matches the Total Transaction Amount card of 153.48bn, so the three measures are internally consistent. Spread over 10,000 accounts, this is about 132 transactions per account.

4.2 Fraudulent transactions



There are 1,719 fraudulent transactions out of about 1.32 million, a fraud rate of approximately 0.13%. This matches the donut chart on the dashboard, which labels non-fraud at 99.87% and fraud at 0.13%. Fraud is therefore rare in relative terms, but at this volume it still represents more than seventeen hundred individual cases that need review.

4.3 Accounts and initial balance

The dataset covers 10,000 accounts with a combined initial balance of 2.80 million, or about 280 per account on average. By comparison, the total transaction amount is 153.48 billion. The money moved is therefore many thousands of times larger than the opening balances, which suggests that funds are recycled repeatedly between accounts or that the amounts are simulated. This should be confirmed with the data owner before drawing financial conclusions.

4.4 Summary of KPIs

KPI	Value	Interpretation
Total Transactions	~1.32 million	High transaction volume; card is truncated on screen.

Average Transaction	115.99K	Consistent with 153.48bn divided by ~1.32M transactions.
Fraudulent Transactions	1,719	Transactions flagged IS_FRAUD = True (about 0.13%).
Total Transaction Amount	153.48bn	Total value moved; currency is not shown.
Total Accounts	10K	About 132 transactions per account on average.
Total Initial Balance	2.80M	About 280 per account; far smaller than amounts moved.

5. Fraud Distribution Analysis

This section analyses the Total Transactions by IS_FRAUD charts (donut and column).



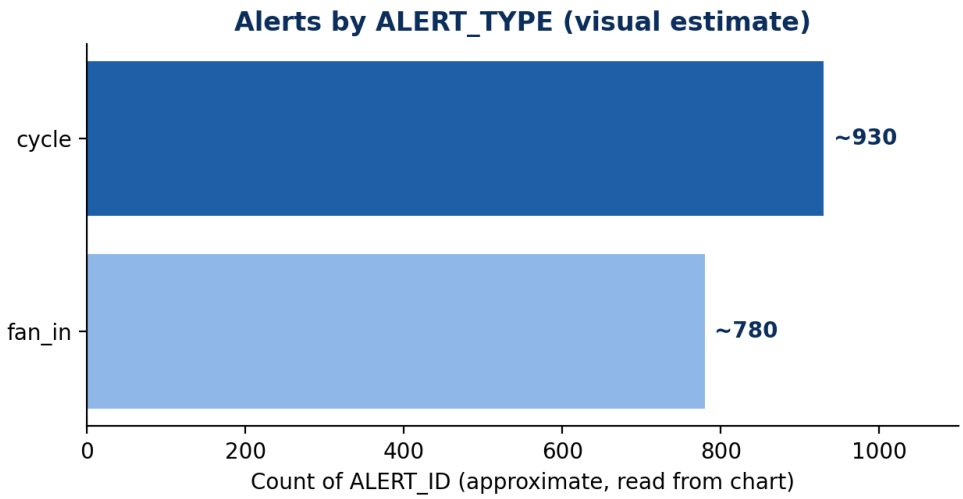
	Share of transactions	Reading
Non-fraud (False)	99.87%	About 1M+ transactions
Fraud (True)	0.13%	1,719 transactions
Total	**100.00%**	**~1.32 million transactions**

- Dominant class: non-fraud transactions make up 99.87% of the data, so the fraud column is barely visible next to the non-fraud column.
- Minority class: fraud accounts for only 0.13%, or about 1 in every 770 transactions.
- Consistency: the donut (proportion) and the column chart (volume) tell the same story from two angles.

Key Insight: *The dataset is extremely imbalanced. For any fraud-detection model or monitoring rule built on this data, plain accuracy would be misleading, because a model that labels everything as non-fraud would still be 99.87% accurate. Precision, recall and F1-score on the fraud class are the measures that matter.*

6. Alert Type Analysis

This section analyses the Count of ALERT_ID by ALERT_TYPE chart.

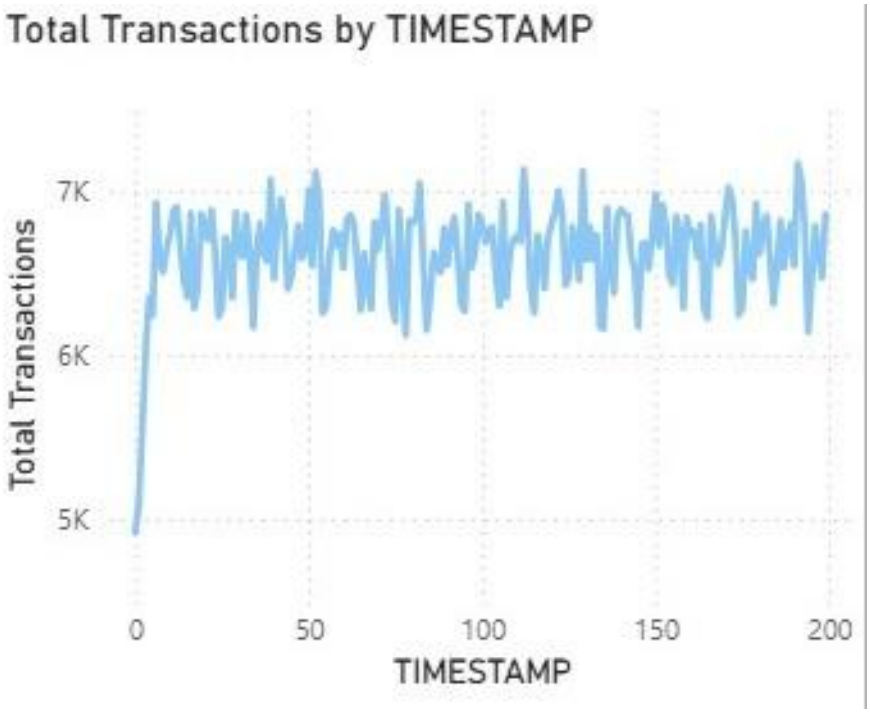


- Most frequent alert: cycle, with roughly 900+ alerts (the bar reaches close to the 1,000 mark).
- Second alert type: fan_in, with roughly 750+ alerts.
- Combined volume: the two alert types add up to roughly 1,700 alerts, which is in the same range as the 1,719 fraudulent transactions.

Key Insight: *Cycle and fan_in are classic money-laundering patterns (funds moving in a loop, or many accounts sending to one account). Cycle alerts are slightly more common. The similarity between the combined alert count and the fraud count suggests the two are closely linked, but the dashboard does not show the mapping, so this should be verified in the data before it is treated as fact.*

7. Transaction Trend Analysis

This section analyses the Total Transactions by TIMESTAMP line chart.



- Ramp-up: transactions start at about 5K per timestamp and rise sharply within the first few steps to about 6.5K.
- Stable band: from then until timestamp 200 the volume stays between roughly 6K and 7.2K with small up-and-down fluctuations.
- No major shock: there is no large spike or collapse in volume at any point on the visible range.
- Tooltip example: at one hovered point (around timestamp 75) the tooltip shows 8 fraudulent transactions against 6,206+ total transactions, a rate of about 0.13%, in line with the overall fraud rate.

Key Insight: *Transaction volume is steady, and the fraud rate at the sampled point matches the overall rate. This suggests fraud is spread across time rather than concentrated in a single burst. However, the chart shows volume only. A separate fraud-by-timestamp view is needed to confirm whether fraud clusters at particular steps.*

8. Business Insights

Each insight below pairs an observation (what the data shows) with the insight (what it means for the business).

Insight 1: Fraud is rare but present at scale

Observation: 1,719 of about 1.32 million transactions (0.13%) are flagged as fraud.

Insight: Because fraud is a tiny minority, monitoring and models must be tuned for rare-event detection. Even a small improvement in detection recovers a meaningful number of cases.

Insight 2: Alert patterns point to specific laundering behaviours

Observation: Cycle alerts (~900+) slightly outnumber fan_in alerts (~750+), and together they roughly match the fraud count.

Insight: Investigation teams can prioritise cycle and fan_in scenarios, and analyse how alerts connect to fraudulent transactions.

Insight 3: Volume is stable over time

Observation: Transactions per timestamp stay in a narrow 6K to 7.2K band after the initial ramp-up, and the sampled fraud rate matches the overall 0.13%.

Insight: Fraud does not appear to be driven by volume spikes, so rules based only on sudden volume changes would miss much of it.

Insight 4: Money moved far exceeds opening balances

Observation: Total transaction amount (153.48bn) is many thousands of times the total initial balance (2.80M).

Insight: Funds are recycled repeatedly or the data is simulated. Either way, the amounts should be checked and the currency confirmed before financial loss is estimated.

9. Business Recommendations

The following recommendations follow directly from the insights above.

- Fraud detection: Evaluate rules and models with precision, recall and F1-score rather than accuracy, and apply imbalance handling such as resampling or class weights.
- Alert review: Prioritise investigation of cycle and fan_in alerts, and verify how each alert maps to the 1,719 fraudulent transactions.
- Dashboard improvement: Add a fraud-by-timestamp view, a fraud-amount KPI and the currency label so the financial impact of fraud is visible.
- Account-level monitoring: Add an account-level view (fraud per account) to find high-risk accounts among the 10K, using the IS_FRAUD slicer to drill in.
- Data quality: Confirm the full value of the truncated Total Transactions card and the tooltip figures, and validate the large gap between initial balances and amounts moved.

10. Conclusion

Overall dashboard findings

The AIML Transactions and Fraud dashboard summarises about 1.32 million transactions worth 153.48 billion across 10,000 accounts. Of these, 1,719 (about 0.13%) are fraudulent. Two alert types, cycle and fan_in, are tracked, and transaction volume stays steady between roughly 6K and 7.2K per timestamp.

Most important business insight

Fraud is a very small share of activity, but it is spread evenly over time and tied to recognisable laundering patterns. The main challenge is therefore detecting a rare class accurately, not reacting to volume spikes.

Key recommendation

Use imbalance-aware metrics for fraud detection, prioritise cycle and fan_in alert reviews, and extend the dashboard with fraud-over-time and fraud-amount views.

How the dashboard helped in the analysis

The KPI cards, donut and column charts, alert chart and trend line turned a large transaction table into a small set of comparable measures. The IS_FRAUD and ALERT_TYPE slicers, together with the tooltip page, make it possible to drill into fraud and alert patterns without leaving the report.